Final Make up Exam

Due: 11:00am on Sunday, July 11, 2021

You will receive no credit for items you complete after the assignment is due. Grading Policy

Q 1

A point charge of -3.00 $\mu\mathrm{C}$ is placed in the center of a spherical cavity of radius 6.90 cm inside an insulating spherical charged solid. The charge density in the solid is 7.35 $\times$ 10^{-4} $\mathrm{C/m^3}$.

Part A

Find the magnitude of the electric field inside the solid at a distance of 9.40 cm from the center of the cavity.

Express your answer with the appropriate units.

ANSWER:

E = $3.528{\cdot}10^{-10}$ $\frac{\mathrm{N}}{\mathrm{C}}$

Completed

Part B

What is the direction of this electric field.

ANSWER:

- radially outward
- radially inward

Completed

Q 2

A flat, square surface with side length 3.45 cm is in the xy-plane at $z = 0$.

Part A

Calculate the magnitude of the flux through this surface produced by a magnetic field $\vec{B} = (\ 0.175\ T)\hat{i} + (\ 0.350\ T)\hat{j} - (\ 0.450\ T)\hat{k}$.

Express your answer in webers.

ANSWER:

$|\Phi_b|$ = $4.6689{\cdot}10^{-4}$ Wb

Completed

Q 3

A wire 30.0 cm long lies along the *z*-axis and carries a current of 8.30 A in the +*z*-direction. The magnetic field is uniform and has components B_x = -0.228 T , B_y = -0.980 T , and B_z = -0.303 T .

Part A

Find the *x*-component of the magnetic force on the wire.

Express your answer in newtons.

ANSWER:

F_x = 0.56772 N

Completed

Part B

Find the *y*-component of the magnetic force on the wire.

Express your answer in newtons.

ANSWER:

F_y = 2.4402 N

Completed

Part C

Find the *z*-component of the magnetic force on the wire.

Express your answer in newtons.

ANSWER:

F_z = 0.75447 N

Completed

Part D

What is the magnitude of the net magnetic force on the wire?

Express your answer in newtons.

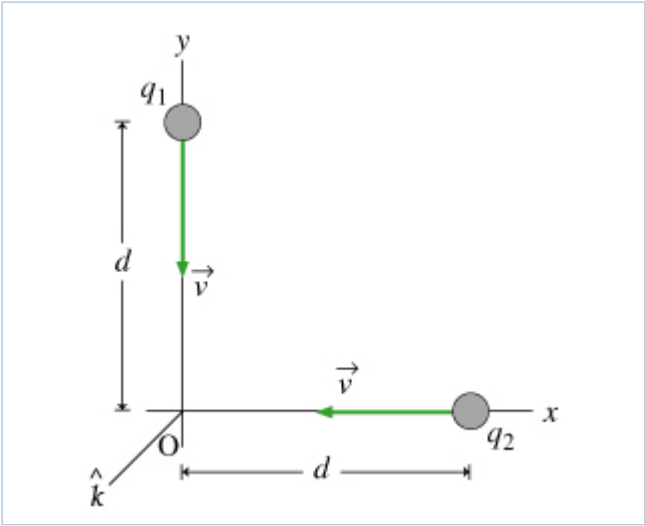
ANSWER:

F = 2.616506 N

Completed

Q 4

Two point charges, with charges q_1 and q_2, are each moving with speed v toward the origin. At the instant shown q_1 is at position (0, d) and q_2 is at (d, 0). (Note that the signs of the charges are not given because they are not needed to determine the magnitude of the forces between the charges.)



Part A

What is the magnitude of the electric force between the two charges?

Express F in terms of q_1, q_2, d, and ϵ_0.

ANSWER:

$F = \dfrac{(q_1 \cdot q_2)}{8\pi\epsilon_0 d}$

Completed

Part B

What is the magnitude of the magnetic force on q_2 due to the magnetic field caused by q_1?

Express the magnitude of the magnetic force in terms of q_1, q_2, v, d, and μ_0.

ANSWER:

$F = \dfrac{(\mu_0 q_1 q_2 \cdot \hat{r} \cdot v^2)}{8\pi d}$

Completed

Part C

Assuming that the charges are moving nonrelativistically ($v \ll c$), what can you say about the relationship between the magnitudes of the magnetic and electrostatic forces?

ANSWER:

- The magnitude of the magnetic force is greater than the magnitude of the electric force.
- The magnitude of the electric force is greater than the magnitude of the magnetic force.
- Both forces have the same magnitude.

Completed

Q 5

Consider that you have a cylindrical copper wire which has 2.10 km length and is connected with a 220.0-V potential difference. (for copper: ρ=1.72x10^{-8} Ω.m)

Part A

What should be its diameter so that it produces heat at a rate of 50.0 W?

Express your answer with the appropriate units.

ANSWER:

D =

Part B

Find the electric field inside the cable under these conditions?

Express your answer with the appropriate units.

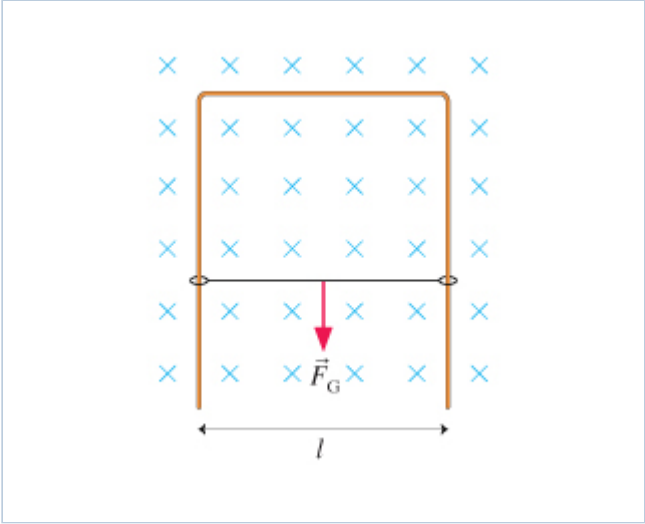
ANSWER:

E = $0\ \frac{\mathrm{N}}{\mathrm{C}}$

Completed

Q 6

The figure shows a U-shaped conducting rail that is oriented vertically in a horizontal magnetic field B. The rail has no electric resistance and does not move. A slide wire with mass m and resistance R can slide up and down without friction while maintaining electrical contact with the rail. The slide wire is released from rest.



Part A

The slide wire reaches a terminal speed v_{term}. Find an expression for v_{term}.

Express your answer in terms of given quantities and appropriate constants.

ANSWER:

v_{term} =

Part B

Determine the value of a terminal velocity v_{term} if l = 23 cm , m = 10 g , R = 0.10 Ω , and B = 0.55 T .

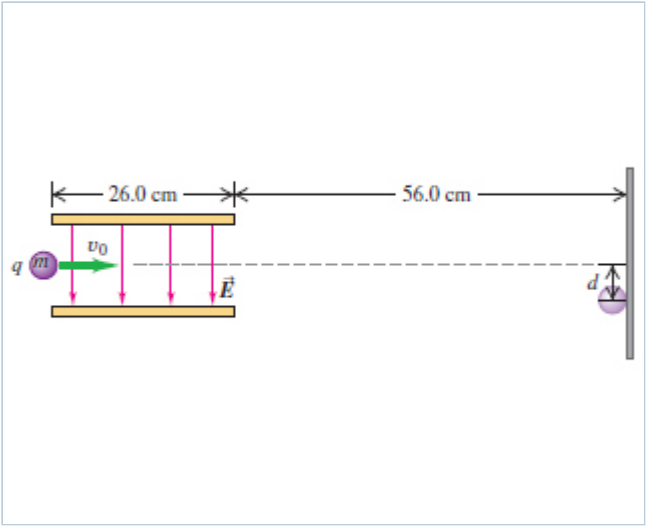
Express your answer to two significant figures and include the appropriate units.

ANSWER:

v_{term} =

Q 7

A small sphere with mass m, charge q, and initial speed v_0 = 5.00×10^3 m/s is projected into a uniform electric field between two parallel metal plates of length 26.0 cm . The electric field between the plates is oriented downward and has magnitude E = 800 N/C . Consider the field is zero outside the region between the plates. The distance between the plates is large enough for the object to pass between the plates without hitting the lower plate. After passing through the field region, the object is deflected downward a vertical distance d = 1.35 cm from its original direction of motion and reaches a collecting plate that is 56.0 cm from the edge of the parallel plates. Ignore gravity and air resistance.



Part A

Find the object's charge-to-mass ratio, q/m.

Express your answer in coulombs per kilogram.

ANSWER:

q/m = $\mathrm{C/kg}$

Q 8

A 2.10 $\mathrm{cm} \times$ 2.10 cm square loop of wire with resistance 1.10×10^{-2} Ω has one edge parallel to a long straight wire. The near edge of the loop is 1.10 cm from the wire. The current in the wire is increasing at the rate of 120 $\mathrm{A/s}$.

Part A

What is the current in the loop?

ANSWER:

$\mu\mathrm{A}$

Q 9

Consider a capacitor which is initially uncharged and connected in series with a resistor and an emf source with $\mathcal{E}$ = 110 V . Just after the circuit is completed, the current through the resistor is 6.6×10^{-5} A. The time constant for the circuit is 5.2 s. (Assume that emf has no internal resistance)

Part A

Calculate the resistance of the resistor?

Express your answer in ohms.

ANSWER:

$R = 9.1820159 \cdot 10^{-9}\ \Omega$

Completed

Part B

Find the capacitance of the capacitor?

Express your answer in farads.

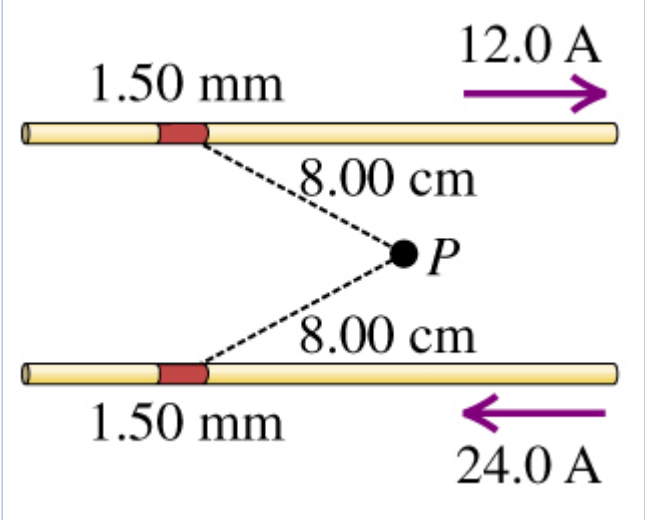
ANSWER:

$C = 6 \cdot 10^{-7}\ \mathrm{F}$

Completed

Q 10

Two parallel wires are 5.60 cm apart and carry currents in opposite directions, as shown in .



Part A

Find the magnitude of the magnetic field at point P due to two 1.50 mm segments of wire that are opposite each other and each 8.00 cm from P.

Express your answer with the appropriate units.

ANSWER:

$B = 25.714 \cdot 10^{-7}\ \mathrm{T}$

Completed

Part B

Find the direction of the magnetic field at point P.

ANSWER:

- (•) into the page
- ( ) out of the page

Completed